

Lec: 15

Eigen value & Eigen vectors

most interesting + highly significant !!

Eigen value decomposition
(EVD)

Consider a 'square matrix'.

$$A \sim n \times n$$

$\bar{u}$ = eigen vector

input $\leftarrow A \bar{u} = \lambda \bar{u}$ $\rightarrow$ output is scaled version of $\bar{u}$.

(corresponding eigenvalue)

$$A \bar{u} = \lambda \bar{u}$$

$$\Rightarrow A \bar{u} - \lambda \bar{u} = 0.$$

$$(A - \lambda I) \bar{u} = 0.$$

$A - \lambda I$ is a singular matrix

non-trivial null space

$$\Rightarrow \boxed{|A - \lambda I| = 0} \rightarrow \text{characteristic equation.}$$

↙ Determinant

solving this eqn. yields eigen values.

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$$

$$A \sim 2 \times 2 \quad A - \lambda I \sim 2 \times 2$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{bmatrix}$$

$$|A - \lambda I| = (2-\lambda)(1-\lambda) - 6 = 0$$

$$\Rightarrow (2-\lambda)(1-\lambda) = 6.$$

$$\Rightarrow 2 - 3\lambda + \lambda^2 - 6 = 0.$$

$$\Rightarrow \lambda^2 - 3\lambda - 4 = 0.$$

$$\Rightarrow (\lambda + 1)(\lambda - 4) = 0$$

$$\Rightarrow \lambda = 4, -1$$

eigen values

To find the eigen vectors,

$$(A - \lambda I) \vec{u} = 0.$$

lies in the null space
of $A - \lambda I$

$$A - \lambda I \quad \lambda = 4$$

$$\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \right) \vec{u} = 0.$$

$$\begin{bmatrix} -2 & 3 \\ 2 & -3 \end{bmatrix} \vec{u} = 0.$$

$$\begin{bmatrix} -2 & 3 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0.$$

$$-2u_1 + 3u_2 = 0$$

$$2u_1 - 3u_2 = 0.$$

$$2u_1 - 3u_2 = 0$$

$$\Rightarrow u_1 = \frac{3}{2} u_2$$

$$u_2 = 2 \Rightarrow u_1 = 3$$

$$\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$\bar{u}$ is an eigen vector

$\Rightarrow \alpha \bar{u}$ is also an eigen vector

$$\overset{A}{\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}} \overset{\bar{u}}{\begin{bmatrix} 3 \\ 2 \end{bmatrix}} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = \overset{\lambda}{4} \overset{\bar{u}}{\begin{bmatrix} 3 \\ 2 \end{bmatrix}}$$

Similarly, $\lambda = -1$

eigen vector $\bar{u}$

$$(A - \lambda I) \bar{u} = 0$$

$$\left(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \right) \bar{u} = 0.$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0.$$

$$\left. \begin{array}{l} 3u_1 + 3u_2 = 0 \\ 2u_1 + 2u_2 = 0 \end{array} \right\} \Rightarrow u_1 = -u_2$$

$$\bar{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A\bar{u} = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\ = (-1) \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\bar{u}}$$

Consider the $n \times n$ matrix A

note $|A - \lambda I| = 0$
 $\rightarrow$ degree = n .

$\Rightarrow$ n distinct roots

$\bar{u}_1, \bar{u}_2, \dots, \bar{u}_n$
 eigen vectors

$\lambda_1, \lambda_2, \dots, \lambda_n$
 distinct eigenvalues

$$A\bar{u}_1 = \lambda_1 \bar{u}_1$$

$$A\bar{u}_2 = \lambda_2 \bar{u}_2$$

$\vdots$

$$A\bar{u}_n = \lambda_n \bar{u}_n$$

$$\begin{bmatrix} A\bar{u}_1 & A\bar{u}_2 & \dots & A\bar{u}_n \end{bmatrix} = \begin{bmatrix} \lambda\bar{u}_1 & \lambda\bar{u}_2 & \dots & \lambda\bar{u}_n \end{bmatrix}$$

$$\Rightarrow \underbrace{A[\bar{u}_1 \ \bar{u}_2 \ \dots \ \bar{u}_n]}_{U - n \times n} = \underbrace{\begin{bmatrix} \bar{u}_1 & \bar{u}_2 & \dots & \bar{u}_n \end{bmatrix}}_U \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}}_{\Lambda}$$

Diagonal
matrix of eigenvalues
 $n \times n$

$$AU = U\Lambda$$

$$A = U\Lambda U^{-1}$$

$$A = U\Lambda U^{-1}$$

diagonal matrix of eigenvalues

EIGENVALUE

DECOMPOSITION

$n \times n$
matrix of
eigenvectors

$$A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \quad \lambda_1 = 4 \quad \lambda_2 = -1$$

$$\overline{u}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \overline{u}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$U = \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}$$

$$U^{-1} = -\frac{1}{5} \begin{bmatrix} -1 & -1 \\ -2 & 3 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$$

$$A = U \Lambda U^{-1}$$

$$= \frac{1}{5} \begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 12 & -1 \\ 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 10 & 15 \\ 10 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$$

